

1. Find the directional derivative of $f(x, y) = \sqrt{4x^2 + 3y}$ at $(2, 3)$ in the direction of $\mathbf{i} - 2\mathbf{j}$.

Proof. Recall that the directional derivative of f in the direction of the unit vector $\mathbf{u}$ is

$$D_{\mathbf{u}}f(x, y) = \nabla f(x, y) \cdot \mathbf{u}.$$

We first compute

$$\nabla f(x, y) = \left\langle \frac{4x}{\sqrt{4x^2 + 3y}}, \frac{3}{2\sqrt{4x^2 + 3y}} \right\rangle.$$

Thus,

$$\nabla f(2, 3) = \left\langle \frac{8}{5}, \frac{3}{10} \right\rangle.$$

The vector $\mathbf{i} - 2\mathbf{j}$ is $\langle 1, -2 \rangle$. Hence, the corresponding unit vector is

$$\mathbf{u} = \frac{\langle 1, -2 \rangle}{\sqrt{1^2 + (-2)^2}} = \left\langle \frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}} \right\rangle.$$

Therefore,

$$\begin{aligned} D_{\mathbf{u}}f(2, 3) &= \nabla f(2, 3) \cdot \mathbf{u} \\ &= \left\langle \frac{8}{5}, \frac{3}{10} \right\rangle \cdot \left\langle \frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}} \right\rangle \\ &= \frac{8}{5\sqrt{5}} - \frac{3}{5\sqrt{5}} \\ &= \frac{1}{\sqrt{5}}. \end{aligned}$$

□

2. Suppose f is a differentiable function of x and y and g is a differentiable function of u and v such that $g(u, v) = f((u + 2v)^3 + 1, e^{uv} - 1)$. Using the given table(see below), find $g_v(-1, 0)$.

Proof. Let $x = (u + 2v)^3 + 1$ and $y = e^{uv} - 1$. By the Chain Rule,

$$g_v(u, v) = f_x(x, y) \frac{\partial x}{\partial v} + f_y(x, y) \frac{\partial y}{\partial v}.$$

We have

$$\frac{\partial x}{\partial v} = 6(u + 2v)^2 \quad \text{and} \quad \frac{\partial y}{\partial v} = ue^{uv}.$$

At $(u, v) = (-1, 0)$, $x = (-1)^3 + 1 = 0$ and $y = e^0 - 1 = 0$. Moreover,

$$\left. \frac{\partial x}{\partial v} \right|_{(-1, 0)} = 6 \quad \text{and} \quad \left. \frac{\partial y}{\partial v} \right|_{(-1, 0)} = -1.$$

From the table, $f_x(0, 0) = 5$ and $f_y(0, 0) = 7$. Therefore,

$$\begin{aligned} g_v(-1, 0) &= f_x(0, 0)(6) + f_y(0, 0)(-1) \\ &= 5(6) - 7 \\ &= 23. \end{aligned}$$

	f	g	f_x	f_y
$(-1,0)$	8	1	4	2
$(0,0)$	1	3	5	7

□